

ATFL311: AUTOMATA THEORY & FORMAL LANGUAGES
Module 2: Finite State Machines & Prerequisites

Comprehensive Identification & Multiple-Choice Reviewer — 48 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (48 ITEMS)

- _____ 1. A computational model that is in exactly one state at any given time, transitioning to a new state based on an incoming input, is a ____ State Machine.
a.) Infinite b.) Turing c.) Finite d.) Pushdown
- _____ 2. In a Finite State Machine, the property of being in ____ state at any given moment distinguishes its operation.
a.) zero b.) exactly one c.) multiple parallel d.) every
- _____ 3. A Finite State Machine changes from its current state to a subsequent state based on an incoming ____.
a.) power surge b.) stack overflow c.) input d.) memory dump
- _____ 4. In formal language theory, an indivisible, atomic unit such as a character or digit (e.g., 'a', 'b', '0', '1') is defined as a ____.
a.) symbol b.) language c.) string d.) grammar
- _____ 5. A finite, non-empty set of symbols is called an ____ and is standardly denoted by the Greek capital letter Sigma (Σ).
a.) transducer b.) language c.) alphabet d.) automaton
- _____ 6. The standard Greek letter used in theoretical computer science to represent an alphabet of symbols is ____.
a.) Epsilon (ϵ) b.) Lambda (λ) c.) Sigma (Σ) d.) Delta (δ)
- _____ 7. A finite sequence formed by concatenating symbols chosen from an alphabet is called a ____.
a.) symbol b.) cardinality c.) language d.) string
- _____ 8. In formal language theory, a collection or set of strings formed over an alphabet Σ is defined as a ____.
a.) language b.) grammar c.) symbol d.) transducer
- _____ 9. If alphabet $\Sigma = \{a, b\}$, the language L_1 containing all strings of length exactly 2 consists of ____ strings.
a.) 16 b.) 4 c.) 2 d.) 8
- _____ 10. If alphabet $\Sigma = \{a, b\}$, which of the following represents the complete set of strings of length 2?
a.) {aaa, aab, aba, abb} b.) {a, b} c.) {aa, ab, ba, bb} d.) {a, b, ab, ba}
- _____ 11. If alphabet $\Sigma = \{a, b\}$, the language L_2 consisting of all strings of length exactly 3 contains a total of ____ strings.
a.) 3 b.) 6 c.) 8 d.) 9
- _____ 12. If alphabet $\Sigma = \{a, b\}$, which of the following strings belongs to the language L_3 defined as 'Set of all strings that begin with a'?
a.) bbb b.) baa c.) bba d.) abb
- _____ 13. The special symbol representing the empty string (a string with a length of zero containing no symbols) is ____.
a.) Phi (ϕ) b.) Epsilon (ϵ) c.) Sigma (Σ) d.) Delta (δ)
- _____ 14. In the powers of alphabet notation, Σ^0 represents the set of all strings of length 0, which is equal to ____.

a.) $\emptyset$ (empty set) b.) $\{0\}$ c.) $\{\epsilon\}$ d.) $\{0, 1\}$

_____ 15. For the binary alphabet $\Sigma = \{0, 1\}$, Σ^1 represents the set of all strings of length 1, which is _____.

a.) $\{01, 10\}$ b.) $\{0, 1\}$ c.) $\{00, 11\}$ d.) $\{\epsilon\}$

_____ 16. For the binary alphabet $\Sigma = \{0, 1\}$, Σ^2 represents the set of all strings of length 2, which is _____.

a.) $\{000, 111\}$ b.) $\{00, 01, 10, 11\}$ c.) $\{0, 1\}$ d.) $\{\epsilon, 01\}$

_____ 17. For the binary alphabet $\Sigma = \{0, 1\}$, Σ^3 represents the set of all strings of length 3, containing _____ distinct strings.

a.) 4 b.) 16 c.) 6 d.) 8

_____ 18. In general formal automata notation, Σ^n represents the set of all strings of length _____ over alphabet Σ .

a.) $n - 1$ b.) $2n$ c.) n d.) n^2

_____ 19. The total number of elements contained in a set is defined in mathematics and automata theory as the set's _____.

a.) permutation b.) transition c.) cardinality d.) closure

_____ 20. For a binary alphabet $\Sigma = \{0, 1\}$, the cardinality of Σ^1 is _____.

a.) 2 b.) 0 c.) 4 d.) 1

_____ 21. For a binary alphabet $\Sigma = \{0, 1\}$, the cardinality of Σ^2 is _____.

a.) 2 b.) 4 c.) 8 d.) 6

_____ 22. For a binary alphabet $\Sigma = \{0, 1\}$, the cardinality of Σ^3 is _____.

a.) 12 b.) 16 c.) 6 d.) 8

_____ 23. For any binary alphabet $\Sigma = \{0, 1\}$, the mathematical formula for the cardinality of Σ^n is _____.

a.) 2^n b.) $n!$ c.) n^2 d.) $2n$

_____ 24. If an alphabet has 3 symbols (such as $\Sigma = \{a, b, c\}$), the cardinality of Σ^2 is _____.

a.) 27 b.) 8 c.) 9 d.) 6

_____ 25. The unary operator that represents the infinite union of all powers of an alphabet ($\Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$) is the _____ star.

a.) Kleene b.) Moore c.) Turing d.) Chomsky

_____ 26. The Kleene closure of an alphabet Σ , denoted Σ^* , always contains the _____ string.

a.) infinite b.) empty c.) undefined d.) negative

_____ 27. For alphabet $\Sigma = \{0, 1\}$, the language defined by Σ^* represents the set of all possible strings of _____ lengths over $\{0, 1\}$.

a.) odd b.) prime c.) even d.) all

_____ 28. The Kleene star operation on an alphabet Σ produces a set that is _____ in size.

a.) strictly finite b.) uncountable c.) empty d.) countably infinite

_____ 29. Finite Automata are primarily divided into two main branches: FA without output and FA _____ output.

a.) beyond b.) with c.) against d.) rejecting

_____ 30. Finite Automata with output include the Mealy Machine and the _____ Machine.

a.) Moore b.) Turing c.) Pushdown d.) Chomsky

_____ 31. In a Moore Machine, the output generated by the automaton depends exclusively on the _____.

a.) tape head position b.) current input symbol c.) current state d.) stack depth

_____ 32. In a Mealy Machine, the output depends on both the current state and the _____.

a.) initial state b.) stack pointer c.) memory bus d.) current input symbol

_____ 33. Which of the following is categorized as a Finite Automaton WITHOUT output (an acceptor/language recognizer)?

a.) DFA b.) Mealy Machine c.) Moore Machine d.) Transducer

_____ 34. The three standard variants of Finite Automata without output are DFA, NFA, and _____-NFA.

a.) ϵ (Epsilon) b.) Σ (Sigma) c.) Q d.) δ (Delta)

- _____ 35. A Finite State Machine is formally defined by how many mathematical components (tuple variables)?
a.) 4 b.) 5 c.) 3 d.) 7
- _____ 36. In the formal 5-tuple definition of an FSM, the variable Q represents the finite set of all _____.
a.) transitions b.) outputs c.) inputs d.) states
- _____ 37. In the formal 5-tuple definition of an FSM, the symbol Σ represents the set of all _____.
a.) state registers b.) inputs (alphabet) c.) transition rules d.) final states
- _____ 38. In the formal 5-tuple definition of an FSM, q_0 denotes the unique _____ state.
a.) start (initial) b.) trap c.) final d.) dead
- _____ 39. In the formal 5-tuple definition of an FSM, F represents the set of all _____ states.
a.) final (accepting) b.) intermediate c.) initial d.) rejected
- _____ 40. In state transition diagrams, an accepting or final state is graphically depicted using a _____ circle.
a.) filled black b.) dotted c.) triangular d.) double
- _____ 41. In state transition diagrams, the start state is graphically indicated by an incoming arrow from _____.
a.) nowhere (outside) b.) the final state c.) a double circle d.) the trap state
- _____ 42. In the formal 5-tuple definition of an FSM, the transition function mapping state-input pairs to a next state is represented by the Greek letter _____.
a.) Delta (δ) b.) Theta (θ) c.) Sigma (Σ) d.) Epsilon (ϵ)
- _____ 43. For a DFA, the mathematical domain and co-domain of the transition function δ is _____.
a.) $Q \times Q \rightarrow \Sigma$ b.) $\Sigma \times \Sigma \rightarrow Q$ c.) $Q \times (\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$ d.) $Q \times \Sigma \rightarrow Q$
- _____ 44. In the Week 2 Slide 8 FSM diagram with states $\{A, B, C, D\}$, start state A, and final state D, what is the destination state from state A on input '0'?
a.) C b.) B c.) D d.) A
- _____ 45. In the Week 2 Slide 8 FSM diagram, what is the destination state from state A on input '1'?
a.) C b.) A c.) B d.) D
- _____ 46. In the Week 2 Slide 8 FSM diagram, transitioning from state B on input '0' leads directly to state _____.
a.) A b.) B c.) D d.) C
- _____ 47. In the Week 2 Slide 8 FSM diagram, transitioning from state C on input '1' leads directly to state _____.
a.) D b.) B c.) A d.) C
- _____ 48. In the Week 2 Slide 8 FSM diagram, if the machine receives the input string '01' starting at state A, the machine terminates in state _____.
a.) D (Accepting) b.) A (Non-accepting) c.) C (Non-accepting) d.) B (Non-accepting)

COMPLETE ANSWER KEY & EXPLANATIONS

Module 2: Finite State Machines & Prerequisites — Reference Key with Theoretical Rationales

Item	Key	Correct Answer	Theoretical Explanation & Concept
1	C	Finite	[Finite State Machine Definition] A Finite State Machine (FSM) is a model of computation that is in exactly one state at any given time and changes state in response to inputs.
2	B	exactly one	[FSM State Exclusivity] At any discrete computational time step, a standard FSM occupies exactly one active state.
3	C	input	[State Transitions] State transitions in an FSM are triggered deterministically or non-deterministically by input symbols provided to the machine.
4	A	symbol	[Symbol Definition] A symbol is an atomic, fundamental entity (letter, digit, or punctuation) that cannot be broken down further.
5	C	alphabet	[Alphabet Definition] An alphabet (Σ) is defined as any finite, non-empty set of symbols (e.g., $\Sigma = \{0, 1\}$ or $\Sigma = \{a, b\}$).
6	C	Sigma (Σ)	[Alphabet Notation] The capital Greek letter Sigma (Σ) is standardly used to denote the input alphabet in automata theory.
7	D	string	[String Definition] A string (or word) is a finite sequence of symbols chosen from a designated alphabet (e.g., '01', 'abb', '1101').
8	A	language	[Language Definition] A formal language L is mathematically defined as a set of strings over an alphabet Σ (i.e., $L \subseteq \Sigma^*$).
9	B	4	[Language of Length 2] For $\Sigma = \{a, b\}$, the strings of length 2 are {aa, ab, ba, bb}, yielding a total of $2^2 = 4$ strings.
10	C	{aa, ab, ba, bb}	[Strings of Length 2 Set] The set of all strings of length 2 over {a, b} is {aa, ab, ba, bb}.
11	C	8	[Language of Length 3] For an alphabet of size 2, the number of strings of length 3 is $2^3 = 8$.
12	D	abb	[Prefix Language Membership] The string 'abb' begins with the symbol 'a', satisfying the language condition L_3 .
13	B	Epsilon (ϵ)	[Empty String Epsilon] Epsilon (ϵ) standardly denotes the empty string, which has a length of zero ($ \epsilon = 0$).
14	C	{ ϵ }	[Sigma Power 0] Σ^0 contains exactly one string—the string of length zero, denoted as { ϵ }.
15	B	{ 0, 1 }	[Sigma Power 1] Σ^1 consists of all single-symbol strings over {0, 1},

			which is $\{0, 1\}$.
16	B	$\{00, 01, 10, 11\}$	[Sigma Power 2] Σ^2 consists of $\{00, 01, 10, 11\}$.
17	D	8	[Sigma Power 3 Count] Σ^3 contains $2^3 = 8$ binary strings: $\{000, 001, 010, 011, 100, 101, 110, 111\}$.
18	C	n	[Sigma Power n] Σ^n is the set of all strings of length n formed from the symbols in alphabet Σ .
19	C	cardinality	[Cardinality Definition] Cardinality (denoted $ S $ or C) is the measure of the number of elements in a set.
20	A	2	[Cardinality of Sigma 1] $\Sigma^1 = \{0, 1\}$, which has a cardinality of 2 elements ($2^1 = 2$).
21	B	4	[Cardinality of Sigma 2] $\Sigma^2 = \{00, 01, 10, 11\}$, which has a cardinality of 4 elements ($2^2 = 4$).
22	D	8	[Cardinality of Sigma 3] Σ^3 has $2^3 = 8$ elements.
23	A	2^n	[Binary Cardinality Formula] Since each position in a string of length n has 2 possible binary choices, the total number of strings is 2^n .
24	C	9	[Ternary Alphabet Cardinality] For an alphabet of size k, the number of strings of length n is k^n . Here $3^2 = 9$.
25	A	Kleene	[Kleene Star Definition] The Kleene star (denoted Σ^*) is the operation that forms the set of all possible strings of all finite lengths over Σ , including ϵ .
26	B	empty	[Kleene Star Contains Epsilon] Because Σ^* includes Σ^0 , it always contains the empty string ϵ .
27	D	all	[Universal String Set] Σ^* is the universal set of all finite strings constructible from the alphabet.
28	D	countably infinite	[Infinite Nature of Kleene Star] Although each individual string in Σ^* is finite in length, the set Σ^* itself contains an infinite number of strings.
29	B	with	[Branches of Finite Automata] Finite Automata divide into FA with output (transducers) and FA without output (acceptors/recognizers).
30	A	Moore	[Moore and Mealy Machines] The two primary finite-state machines with output are Moore machines and Mealy machines.
31	C	current state	[Moore Machine Output Logic] In a Moore machine, output values are determined solely by the current state.
32	D	current input symbol	[Mealy Machine Output Logic] In a Mealy machine, output values are

			determined by both the current state and the current input symbol.
33	A	DFA	[FA Without Output Category] DFAs, NFAs, and ϵ -NFAs are language recognizers (acceptors) that do not generate output sequences beyond accepting or rejecting.
34	A	ϵ (Epsilon)	[Three FA Acceptor Types] The three non-output finite automata are Deterministic (DFA), Non-deterministic (NFA), and NFA with Epsilon transitions (ϵ -NFA).
35	B	5	[5-Tuple Definition] Both DFAs and NFAs are formally defined as 5-tuples: $(Q, \Sigma, q_0, F, \delta)$.
36	D	states	[Variable Q: States] Q represents the finite non-empty set of internal states of the machine.
37	B	inputs (alphabet)	[Variable Sigma: Alphabet] Σ represents the finite input alphabet.
38	A	start (initial)	[Variable q0: Start State] $q_0 \in Q$ represents the initial state where computation begins.
39	A	final (accepting)	[Variable F: Final States] $F \subseteq Q$ represents the set of accept or final states.
40	D	double	[Double Circle Graphical Notation] In standard automata transition diagrams, accepting/final states are indicated by concentric double circles.
41	A	nowhere (outside)	[Start State Arrow Notation] The initial start state is distinguished by an unlabelled arrow entering it from outside the graph.
42	A	Delta (δ)	[Transition Function Delta (δ)] The lowercase Greek letter Delta (δ) standardly designates the state transition function.
43	D	$Q \times \Sigma \rightarrow Q$	[DFA Transition Function Signature] For a DFA, δ maps a state and an input symbol deterministically to exactly one state: $\delta: Q \times \Sigma \rightarrow Q$.
44	A	C	[Slide 8 Transition from A on 0] In the Slide 8 transition diagram, the transition from state A labeled with 0 points directly down to state C.
45	C	B	[Slide 8 Transition from A on 1] In the Slide 8 transition diagram, the horizontal transition from state A labeled with 1 points to state B.
46	C	D	[Slide 8 Transition from B on 0] In the Slide 8 transition diagram, state B on input 0 transitions to state D.
47	A	D	[Slide 8 Transition from C on 1] In the Slide 8 transition diagram, state C on input 1 transitions to state D.
48	A	D (Accepting)	[Slide 8 String Trace '01'] From A, on 0 the machine moves to C. From C, on 1 the machine moves to D. Since D is a final state, string '01' is

accepted.
